

Slides will be provided, but it's also possible to study from books, like "Kurose and Ross, Computer Networking: a Top Down Approach".

Introduction

Recap of Concepts

If we want to model a network, we can do that with a definition of a graph, which is a set of nodes and interconnections between nodes. We can also define a graph as a matrix, a matrix of zero and ones telling if there is connection between node i and j .

Internet is a network of networks, consisting of end systems connected through each other between routers.

A node delay is the time necessary for the packet to go from a source to a destination, and come back, also known as Round Trip Time (RTT). It does not specify anything about the technology, the channel, the medium. So, when dealing about delay, we need to specify queueing delaying, processing delay, etc., and the sum of all these components is the RTT.

When talking about propagation delay we're dealing about the space over speed, where the space is the space it takes for a packet to go from a source to a destination.

We're talking about delays as it's important when we'll deal with Wi-Fi technology, as it's important to understand if the propagation delay has an impact on collision probabilities. The transmission delay, instead, is given by the length of the packet over the channel capacity, bytes/bits per second. The queueing delay is the time the packet remains in a buffer to be processed and delivered. When the traffic loads, the waiting time increases as well.

The throughput depends on each hop that the packet passes through, and if in a given hop we have a decrease of performance, the end to end throughput is the minimum between everything.

Then, we have the difference between Client Server architecture and Peer to Peer architecture. *Dive deeper.*

Medium Access Protocol

We start with the definition of multi-point networks, we can see two different main topologies:

- Start topology
- Bus topology

Then, we'll see several techniques for access protocols:

- Static
 - FDMA
 - TDMA
 - CDMA
- Dynamic
 - ALOHA
 - CSMA

It's a technique providing rules how a node can transmit over a medium, as a medium is not dedicated to one user, rather to multiple ones. If we transmit all together at the same time, we'll have collisions, due to interferences with other signals. Thus, the receiver will not be able to distinguish data that the source 1 is sending, from the data sent from another source 2.

Communication channels allows remote data transfer, and is the physical medium.

We have different transmission modes:

- Point to Point: one source, one destination.

There is not an ideal channel, there will always be some noise, and if the signal to noise ratio is small, the quality will decrease. To measure quality, we can distinguish between quality of service (QoS) and quality of experience. The former is measured by means of delay, throughput, packet loss, jitter. So it depends essentially on SNR , and sometimes SIR (??).

A multipoint network has limited geographical extent, and is referred as LAN. We can define it according to an access protocol and a network topology. As we said, for network topologies we can have busses, rings, star. A common characteristic of all wired and wireless network topology is the broadcast nature of transmission. How do we distinguish different access technologies? We have static and dynamic ones.

For static, it means that there is a channel assignment, so we can divide the capacity of the network according to parameters. IN case of TDMA, we can have the organization in different time slots, in which the user will be able to transmit (pre-assigned time slot). For dynamic access techniques we take into account the

needs for transmission, so that the access is granted based on the request to it. It's more efficient, but also more complex, as we have to consider a protocol that, according to the needs of user, allows the user to transmit or not. Packet collision is not necessarily avoided, and so we have to find some solutions to reduce that.

Channel assignment techniques: We have FDMA, TDMA and CDMA. In each of these, we can see that users will have the information about the time slot he can transmit (with TDMA), or the sub-band (FDMA). In CDMA, a given user has a code, orthogonal to all the other codes used by other users. Each user can transmit simultaneously with this.

In FDMA, each station can access the transmission medium using a pre-assigned band. In TDMA, instead, the whole capacity is made available. Since there is a dependance on time, we need to provide a synchronization technique.

The CDMA makes the entire bandwidth available at all stations, which can transmit simultaneously.

For dynamic access techniques, we consider the needs of the station. If a station wants to transmit, it has to make a request. We can have centralised or distributed techniques. For the former, we have a centralised device which receives request for access to the medium, and provides authorisation. We have mono-directional channels and bi-directional, used to communicate the sequence of access. For distributed access techniques, the stations mutually coordinate to each other. This is achieved using a single bi-directional channel.

These access mechanisms belong to the resource allocation issue. When we talk about resource in static approach, it's the time slot, or sub-band, etc, while for the dynamic the medium is a single shared resource.

Let us consider N stations on a channel. What is the probability a packet will be generated in a given instant? It's

$$\lambda \Delta T$$

λ is the arrival rate (how many packets arrive in a second), multiplied by the time interval.

Once the packet has been generated, the station will be blocked until the packet has been transmitted. If there are multiple transmissions, a collision can occur.

What are performance KPI (key performance indicator)? We talk about throughput, time delay and packet loss probability.

We can define the traffic offer to the network:

$$A_0 = N \cdot \lambda \cdot L$$

N is the number of nodes, λ is the arrival rate (average), L is the packet size. The throughput is the portion of the offered traffic successfully transferred

$$A_s = N\lambda L \cdot (1 - \Pi_p)$$

$(1 - \Pi_p)$ is the successful probability of transmission.

In case of no packet loss, then $\Pi_p = 0$, and it's possible to express the throughput over the channel capacity, normalised.

$$S = \frac{N \cdot \lambda \cdot L}{C}$$

where S is the network utilisation coefficient, and C is the capacity. If the ratio is close to 1, then the network is used fully, and so resources are being used by different users.

Then, the network throughput is given by $S \cdot C$.

In TDMA we can define a time slot sufficient to transmit one information unit (packet), allocated to each station. By having N users, we'll have to define a frame, comprised by multiple time slots. The time access will be divided in fixed duration frames. The frame s will then have a duration given by $\frac{L}{C}$. This frame will be repeated every time. Each user will have to know when to transmit, and will have to wait for its time slot. As we said, this requires synchronization.

The network throughput is given by the sum of the traffic received by all stations. The transmission delay is $\frac{L}{C}$, then we can express the throughput as:

$$S = \rho = N \cdot \lambda T_t$$

We assume zero propagation delay.

The UI will be characterised then by:

- transmission time
- waiting time (queueing time)
- waiting time due to the occurrence of the time slot assigned to me

Thus, the delay in TDMA is given by:

$$D_{TDMA} = \frac{L}{C} + \frac{\rho}{2(1 - \rho)} \frac{NL}{C} + \frac{1}{2} \cdot \frac{NL}{C}$$

The second term is the queueing time in the buffer, and the last is the slot waiting time.

In FDMA, we divide the channel frequency, We thus have different subbands, given by $\frac{C}{N}$. We want to avoid aliasing effect, given by the overlapping of another bandwidth, as it provides interferences.

We'll thus have some guard bands to limit interferences between adjacent channels.

We can also distinguish between FDD (Frequency Division Duplex), through which we provide bi-directional communication, and we allocate a group of sub-bands for uplink, and a group for downlinks. Each user will thus have a pair of sub-bands.

In TDD (time division duplex), each channels becomes bidirectional. We don't have a sub-band for uplink and one for downlink.

In this case, we need to consider synchronization. The advantage is that we're not considering two separate bandwidths.

In FDMA, the spectrum is divided into equal portions assigned to individual stations. By having N stations, we'll have $\frac{C}{N}$ sub-bands. The transmission delay is thus given by:

$$T_{FDMA} = \frac{L}{\frac{C}{N}} = \frac{NL}{C}$$

So, it's N times slower than TDMA.

The throughput is thus given by:

$$S = \rho = \lambda T_{FDMA} = \lambda \cdot \frac{NL}{C}$$

It does not change with respect to TDMA.

What will be different is the delay, the time necessary to achieve that throughput.

$$D_{FDMA} = \frac{NL}{C} + \frac{\rho}{2(1-\rho)} \frac{NL}{C}$$

We don't have the third time, which was the time necessary to wait for the time slot occurrence. This is faster, however we have an increase on transmission delay.

We can normalize everything by dividing for $\frac{L}{C}$, and see how the delay for TDMA is greated by $(\frac{N}{2} - 1)$ than FDMA.

How do we model this normalised delay? We do it according to the throughput. The average normalised delay will be smaller at first with TDMA, but then converge with the one of FDMA as the throughput grows.